MIL-P-82427C(OS)
13 MARCH 1985
SUPERSEDING
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2 May 1969 and
Amendment 3
13 May 1974

MILITARY SPECIFICATION

POWER UNIT
(For Torpedo Mk 46 Mod 1)

This specification is approved for use by the Naval Sea Systems Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of two types of power units (see 6.1.2).

1.2 Classification. The power units covered by this specification shall be of the following types, as specified (see 6.2):

a. Type I shall be in accordance with DL 2539015.

b. Type II shall be in accordance with DL 2557500.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-82426	Torpedo Main Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be address to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

2.1.2 Other Government documents, drawings and publications. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL 1330425	Power Unit, Shipping (Packing) Condition and all documents listed thereon
DL 2539015	Power Unit, and all documents listed thereon
DL 2557500	Power Unit, and all documents listed thereon
2539016	Housing Module No. 7 Power Unit
2539026	Control Group
2539311	Nose Section
2540730	Power Unit Test Set
2557511	Housing, Power Controller

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Basic requirements.

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3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract, all materials and treatments of materials and parts used in manfacture of the power unit shall be in accordance with DL 2539015 or DL 2557500 (See 6.2).

3.2.2 Unless otherwise specified, all requirements of this specification apply both to Type I and Type II power units.

3.3 Performance requirements and product characteristics.

NOTE: The Column I numerical values of this specification shall be applicable to the initial quality conformance tests of 4.7 performed on a unit after its original calibration or after a later recalibration. The Column IA values shall be applicable to all subsequent quality conformance tests except those tests where other values are specified. The Column IB values shall be applicable when specified in TABLE V or TABLE VI. The Column II values shall be applicable to the tests of 4.5.2.4.2 and when specified in TABLE V and TABLE VI. When testing for conformance to Column IA, IB, or II requirements, those parameters for which values are not specified shall meet values specified for Column I. Minimum or maximum values apply to absolute values only (i.e., -15.6 volts direct current (vdc) is greater than -15.1 vdc).

3.3.1 Inputs. The following inputs shall be provided to power units:

3.3.1.1 Input d-c power requirements. Input dc power requirements to power unit shall be as follows:

	Column I	Column II
a. Voltage requirements:		
1. Power supply section		
+40 vdc	$+40.0\ ^{+6.0}_{-4.0}$ vdc	$+40.0\ ^{+6.0}_{-4.0}$ vdc
-40 vdc	$-40.0\ ^{+4.0}_{-6.0}$ vdc	$-40.0\ ^{+4.0}_{-6.0}$ vdc
2. Actuator controller section		
+40 vdc	$+40.0 \pm 0.5$ vdc	$+40 \pm 1$ vdc
-40 vdc	-40.0 ± 0.5 vdc	-40 ± 1 vdc
b. Transient requirements:		
Maximum transient voltage fluctuation from the average absolute value of +40 vdc or the -40 vdc sources.	None	None

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	Column I	Column II
c. Current requirements:		
1. Power supply section:		
+40 vdc	0.5 to 3.0 amperes (amps)	0.5 to 4.0 amps
-40 vdc	1.5 to 5.0 amps	2.0 to 6.0 amps
Additional pulse current for exploder arming (3.3.2.3) from each 40 vdc supply	2.0 to 2.7 amps	1.0 to 3.3 amps
2. Actuator controller section:		
+40 vdc	1.05 to 2.35 amps	0.5 to 2.9 amps
-40 vdc	0.95 to 2.25 amps	0.4 to 2.8 amps

3.3.1.2 Signal inputs.

COMPUTER ENABLE		
Activated	-15.0 ± 0.5 vdc	-15.0 ± 0.8 vdc
Not activated	0.00 ± 0.25 vdc	0.00 ± 0.25 vdc
PITCH DRIVE		
Up Command	-1.0 to -2.4 vdc	-0.75 to -2.75 vdc
Down Command	30 k to 47 k to GRD.	30 k to 47 k to GRD.
YAW DRIVE		
Left Command	-1.0 to -2.4 vdc	-0.75 to 2.75 vdc
Right Command	30 k to 47 k to GRD.	30 k to 47 k to GRD.
ROLL DRIVE		
CCW Command	-1.0 to -2.4 vdc	-0.75 to -2.75 vdc
CW Command	30 k to 47 k to GRD.	30 k to 47 k to GRD.

3.3.2 Power Outputs.

NOTE: Tolerances and ranges for Columns I. IA. IB, and II requirements are tabulated in TABLE I.

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3.3.2.1 DC power outputs. The power unit shall supply the following dc output voltages at the specified load currents:

Load	Voltage
120 to 220 ma (milliamperes)	+30 VDC
170 to 295 ma	+15 VDC
20 to 70 ma	+5 VDC
365 to 569 ma	-30 VDC GUIDANCE
190 to 240 ma	-30 VDC
925 to 1075 ma	-15 VDC
0 ma	+90 VDC IN (2 outputs)

3.3.2.2 AC output power. The power unit shall supply an ac output with the following charactertics at the specified load:

Output characteristics	Value
Load Impedance (capacitive)	60 / -12° to 100 / -34° ohms
Voltage	26.0 volts root mean square (vrms)
Frequency	4800 Hz
Distortion	3.0% maximum

3.3.2.3 Exploder arm output. With the EXPLODER ARM output connected to the ciruit shown in FIGURE 1, the output voltage shall be a funtion of the COMPUTER ENABLE input.

a. COMPUTER ENABLE not activated: EXPLODER ARM output shall exhibit a voltage that, when subtracted from +40 vdc, shall be 0.4 vdc maximum.

b. COMPUTER ENABLE activated: EXPLODER ARM output shall exhibit a voltage that, when subtracted from -40 vdc, shall be 1.5 mazimum.

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TABLE I

Requirements, outputs, and tolerances

Paragraph	Function	Nominal Value	Tolerance column I	IA	IB	II
3.3.2.1	DC outputs	+30.0 vdc	±0.6	±0.7	±0.8	±1.5
		+15.0 vdc	±0.5	±0.6	±0.7	±0.8
		+5.00 vdc	±0.25	±0.25	±0.25	±0.25
		-30.0 vdc	-0.9, +1.1	-0.95, +1.15	-1.00, +1.20	-1.6, +2.1
		Guidance	-1.20	-1.25	-1.30	
		-30.0 vdc	+ .50	+0.55	+0.60	±1.9
		-15.0 vdc	±0.5	±0.6	±0.7	±0.8
		+95.0 vdc	±5.0	±6.0	±7.0	±12.0
		(max overshoot for approx 0.5 secs VDC)	+50	+50	+50	+60
3.3.2.2	AC output:Voltage	26.0 vrms	±0.5	±0.7	±1.0	±2.0
	Frequency	4800 Hz	±30	±35	±50	±100
	Distortion	(% max)	3.0	3.0	3.1	3.2
3.3.2.3(a)	Computer off	Diff. voltage 0.4 vdc max	0.4 max	0.4 max	0.4 max	0.6 max
(b)	Computer on	Diff. voltage 1.5 vdc max	1.5 max	1.5 max	1.5 max	1.6 max
(c)	Without Resistor	variance 0.5 vdc max	0.5 max	0.5 max	0.5 max	0.5 max
3.3.2.4(a)	Alternator	1.5 vdc	0.0 to 1.5	0.0 to 2.0	0.0 to 3.0	0.0 to 20.0
(b)	Regulator	+35.0 vdc	30.0 to 40.0	29.0 to 40.0	27.0 to 40.0	20.0 to 40.0
(c)		-24.0 vdc	-6.0 to -42.0	-6.0 to -42.0	-5.0 to -42.0	-2.0 to -45.0
(d)		-40.6 vdc	±0.6	±1.0	±1.5	±2.0
3.3.3.1.1(a)	Elev. up (-)	-35.5 vdc	±1.5	±1.6	±1.7	±2.0
(b)		-0.05 vdc	-0.2 to +0.1	-0.3 to +0.1	-0.4 to +0.1	-1.0 to +0.1

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TABLE I (Continued)

Requirements, outputs, and tolerances

Paragraph	Function	Nominal Value	Tolerance column I	IA	IB	II
3.3.3.1.2(a)	Elev. down (+)	+0.05 vdc	-0.1 to +0.2	-0.1 to +0.3	-0.1 to +0.4	-0.1 to +1.0
(b)		+35.5 vdc	±1.5	±1.6	±1.7	±2.0
3.3.3.1.3(a)	Upper right (-)	-23.8 vdc	±2.0	±2.1	±2.3	+3.0, -3.4
(b)		-35.5 vdc	±1.5	±1.6	±1.7	±2.0
(c)		-0.05 vdc	-0.2 to +0.1	-0.3 to +0.1	-0.4 to +0.1	-1.0 to +0.1
(d)		-0.05 vdc	-0.2 to +0.1	-0.3 to +0.1	-0.4 to +0.1	-1.0 to +0.1
3.3.3.1.4(a)	Upper left (+)	+0.05 vdc	-0.1 to +0.2	-0.1 to +0.3	-0.1 to +0.4	-0.1 to +1.0
(b)		+0.05 vdc	-0.1 to +0.2	-0.1 to +0.3	-0.1 to +0.4	-0.1 to +1.0
(c)		+35.5 vdc	±1.5	±1.6	±1.7	±2.0
(d)		+23.8 vdc	±2.0	±2.1	±2.3	±4.8, -3.0
3.3.3.1.5(a)	Lower right (+)	+35.5 vdc	±1.5	±1.6	±1.7	±2.0
(b)		+23.8 vdc	±2.0	±2.1	±2.3	±4.8, -3.0
(c)		+0.05 vdc	-0.1 to +0.2	-0.1 to +0.3	-0.1 to +0.3	-0.1 to +1.0
(d)		+0.05 vdc	-0.1 to +0.2	-0.1 to +0.3	-0.1 to +0.4	-0.1 to +1.0
3.3.3.1.6(a)	Lower left (-)	-0.05 vdc	-0.2 to +0.1	-0.3 to +0.1	-0.4 to +0.1	-0.1 to +0.1
(b)		-0.05 vdc	-0.2 to +0.1	-0.3 to +0.1	-0.4 to +0.1	-0.1 to +0.1
(c)		-23.8 vdc	±2.0	±2.1	±2.3	+3.0, -3.4
(d)		-35.5 vdc	±1.5	±1.6	±1.7	±2.0
3.3.3.2	Pulsed outputs:					
	Voltage (-)	-20.5 vdc	±4.5	±4.8	±5.0	±5.5
	Fall time (-)	usec	50 max	55 max	60 max	65 max
	Voltage (+)	+20.5	±4.5	±4.8	±5.0	±5.5
	Fall time (+)	usec	25 max	27 max	28 max	30 max

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TABLE I (Continued

Requirements, outputs, and tolerances

Paragraph	Function	Nominal Value	I	Tolerance column IA	IB	II
3.3.3.3(a)	Cont. oscillator	-1.1 vdc max	-1.1 max	-1.1 max	-1.2 max	-1.4 max
(b)		-23.5 vdc	±1.0	±1.3	±1.5	±1.9
(c)		2.2 ms	±0.4	±0.45	±0.5	±0.6
(d)		67%	±3.0	±3.5	±4.0	±5.0
3.3.3.4	Pitch inhibit:					
(a)	Time	2.0 sec	±0.2	±0.25	±0.3	±0.4
(b)	Not activated	-24.0 vdc	±2.0	±2.2	±2.5	±3.0
(c)	Activated	-1.0 vdc max	-1.0 max	-1.1 max	-1.2 max	-1.4 max
3.3.3.5(a)	Low voltage	-26.0 vdc	±3.0	±3.2	±4.0	±6.0
(b)	Circuit	-25.5 vdc	±3.0	±3.2	±4.0	±6.0

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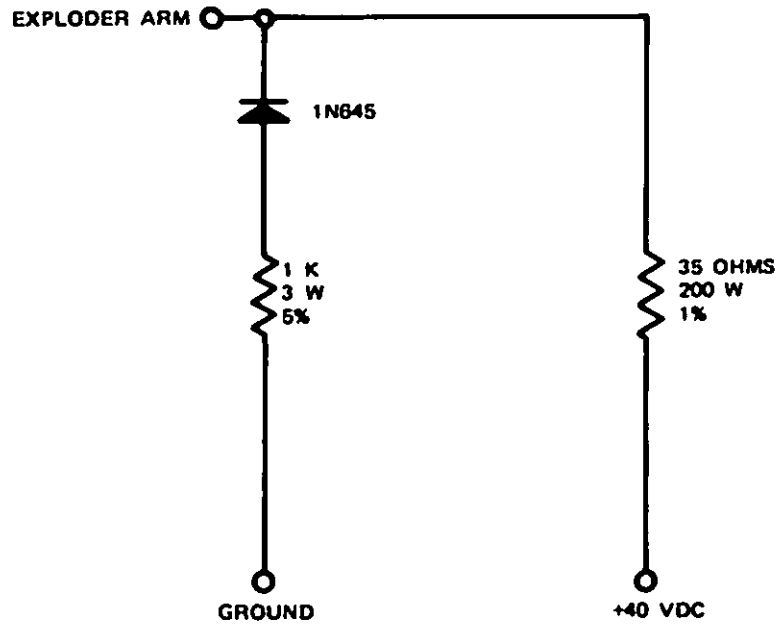


FIGURE 1. Exploder arm circuit

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c. When the 25 ohm resistor is removed, the EXPLODER ARM voltage shall not vary more than 0.5 vdc.

3.3.2.4 Alternator regulator output. The power unit shall provide a SHUNT FIELD output voltage that is a function of the voltages supplied to the +40 vdc and -40 vdc inputs. With a 20 ohm, 1 percent, 150 watt resistor connected between the SHUNT FIELD and FIELD RETURN terminals, the output voltage shall be as specified below:

NOTE: The 20 ohm resistor shall be connected only during tests of this function. The unit should be operated for about 15 minutes, or until thermal equilibrium, without shunt field load connected.

a. With the input voltage at 45.0 +0.2 -0.1 vdc and -37.5 ±0.5 vdc, the SHUNT FIELD voltage shall be +1.5 vdc maximum.

b. With the input voltages at +40.0 to ±0.5 vdc and -37.5 ±0.5 vdc, the SHUNT FIELD voltage shall be +35.0 vdc.

c. With the input voltage at +40.0 ±0.5 vdc and -42.5 ±0.5 vdc, the SHUNT FIELD voltage shall be -24.0 vdc.

d. With the +40 vdc input voltage at +40.0 ±0.5 vdc and the -40 vdc input adjusted for a SHUNT FIELD voltage of 0.0 ±0.2 vdc, the -40 vdc input shall be -40.6 vdc.

3.3.3 Controller outputs.

3.3.3.1 Steady state steering outputs. The power unit shall provide six steering outputs that respond to the PITCH DRIVE, YAW DRIVE, and ROLL DRIVE signal inputs (3.3.1.2). With a 50 ±4 ohm, 50 watt resistor connected from each of the six steering outputs to ground, the following ouput voltages shall exist for the specified conditions:

3.3.3.1.1 Elev Up (-) output.

Input	Steering output
a. PITCH DRIVE (Up Command)	-35.5 vdc
b. PITCH DRIVE (Down Command)	-0.05 vdc

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3.3.3.1.2 Elev Down (+) output.

Input	Steering output
a. PITCH DRIVE (Up Command)	+0.05 vdc
b. PITCH DRIVE (Down Command)	+35.5 vdc

3.3.3.1.3 UPPER RIGHT (-) output.

Input	Steering output
a. YAW DRIVE (Right Command) ROLL DRIVE (CW Command)	-23.8 vdc (average output. Actual output will pulse as specified in 3.3.3.3)
b. YAW DRIVE (Right Command) ROLL DRIVE (CCW Command)	-35.5 vdc
c. YAW DRIVE (Left Command) ROLL DRIVE (CW Command)	-0.05 vdc
d. YAW DRIVE (Left Command) ROLL DRIVE (CCW Command)	-0.05 to vdc

3.3.3.1.4 UPPER LEFT (+) output.

Input	Steering output
a. YAW DRIVE (Right Command) ROLL DRIVE (CW Command)	+0.05 vdc
b. YAW DRIVE (Right Command) ROLL DRIVE (CCW Command)	+0.05 vdc
c. YAW DRIVE (Left Command) ROLL DRIVE (CW Command)	+35.5 vdc
d. YAW DRIVE (Left Command) ROLL DRIVE (CCW Command)	+23.8 vdc (Average output. Actual output will pulse as specified in 3.3.3.3)

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3.3.3.1.5 LOWER RIGHT (+) output.

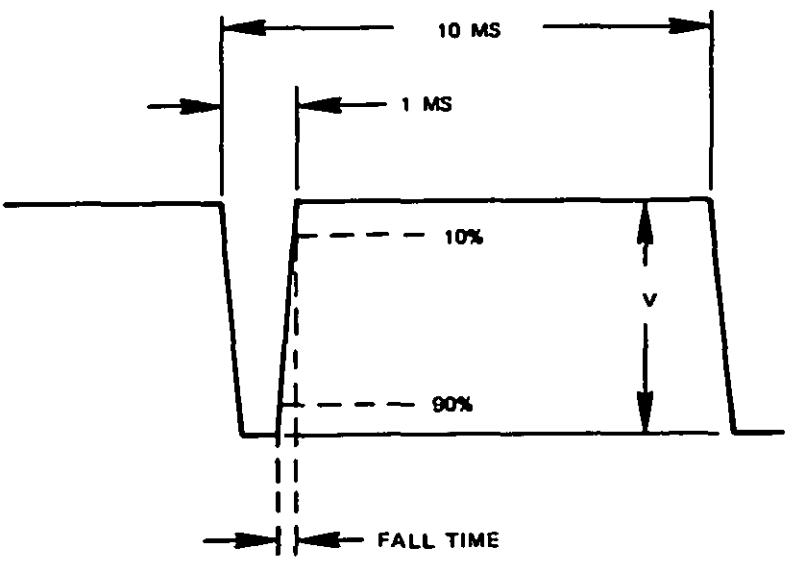
Input	Steering output
a. YAW DRIVE (Right Command) ROLL DRIVE (CW Command)	+35.5 vdc
b. YAW DRIVE (Right Command) ROLL DRIVE (CCW Command)	+23.8 vdc (Average output. Actual output will pulse as specified in 3.3.3.3)
c. YAW DRIVE (Left Command) ROLL DRIVE (CW Command)	+0.05 vdc
d. YAW DRIVE (Left Command) ROLL DRIVE (CCW Command)	+0.05 vdc

3.3.3.1.6 LOWER LEFT (-) output.

Input	Steering output
a. YAW DRIVE (Right Command) ROLL DRIVE (CW Command)	-0.05 vdc
b. YAW DRIVE (Right Command) ROLL DRIVE (CCW Command)	-0.05 vdc
c. YAW DRIVE (Left Command) ROLL DRIVE (CW Command)	-23.8 vdc (Average output. Actual output will pulse as specified in 3.3.3.3)
d. YAW DRIVE (Left Command) ROLL DRIVE (CCW Command)	-35.5 vdc

3.3.3.2 Pulsed steering output. With a pulsed load resulting in approximately 3.8 amperes on each of the six steering outputs, the output voltage pulses shall be as specified below. The pulsed loads shall have a duration of 2 milliseconds (ms) maximum and a duty cycle of 10 percent maximum (see FIGURE 2)

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NOTE: PULSE AMPLITUDE WILL BE POSITIVE AT FUNCTION POSITIONS 1, 4, AND 6 AND NEGATIVE AT FUNCTION POSITIONS 2, 3, AND 5.

FIGURE 2. Pulse current waveform

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Pulsed steering output	Value
ELEV UP (-) Amplitude	-20.5 vdc
Fall time	50 microseconds (μ sec) maximum
ELEV DOWN (+) Amplitude	+20.5 vdc
Fall time	25 μ sec maximum
UPPER RIGHT (-) Amplitude	-20.5 vdc
Fall time	50 μ sec maximum
UPPER LEFT (+) Amplitude	+20.5 vdc
Fall time	25 μ sec maximum
LOWER RIGHT (+) Amplitude	+20.5 vdc
Fall time	25 μ sec maximum
LOWER LEFT (-) Amplitude	-20.5 vdc
Fall time	50 μ sec maximum

3.3.3.3 Controller oscillator. The voltage waveform appearing at test TP1701 shall be as defined in FIGURE 3 and shall meet the following specifications:

Waveform characteristics	Value
a. V1	-1.1 vdc maximum
b. V2	-23.5 vdc
c. T1	2.2 vdc
d. T2/T1	67 vdc

3.3.3.4 Pitch inhibit. The power unit shall supply a ground to the pitch inhibit terminals at a specified time after application of -40 vdc input voltage.

Characteristics	Value
a. Time	2.0 seconds
b. Nonactivated voltage	-24.0 vdc
c. Activated voltage	-1.0 vdc maximum

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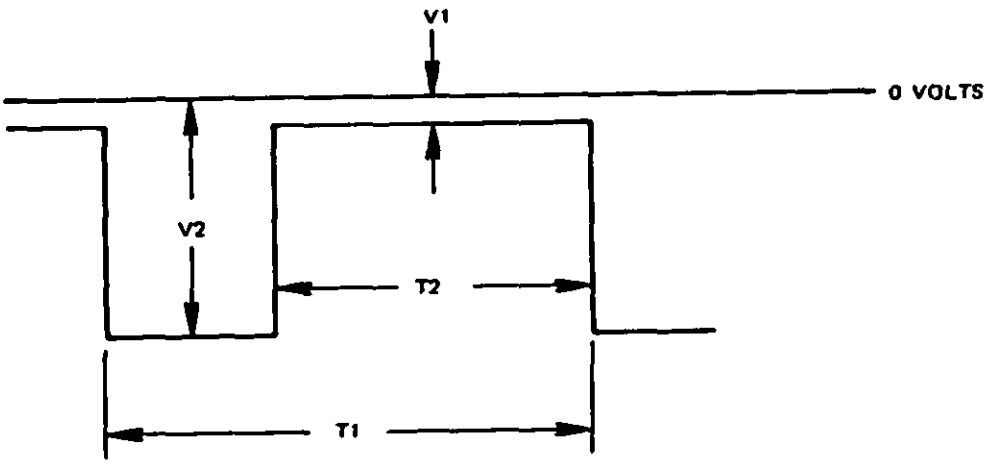


FIGURE 3. Controller oscillator waveform

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3.3.3.5 Low voltage circuit.

a. The power unit shall inhibit the six steering outputs until the voltage at -40 VDC terminals reaches -26.0 vdc.

b. The power unit shall inhibit the six steering outputs when the voltage at the -40 VDC terminals drops to -25.5 vdc.

3.3.4 Electrical continuity and isolation.

3.3.4.1 The resistance between pin UU of P1501 and any combination of pins F, H, J, U, Y, SS and VV of P1501 shall be 0.5 ohm maximum.

3.3.4.2 The resistance between the power unit casting and pins A, B, R, S, T, HH, TT, UU, WW and XX of P1501 shall be greater than 500,000 ohms.

3.3.4.3 The resistance between pins HH and XX of P1501 shall be greater than 500 ohms.

3.3.4.4 The resistance between pins TT and WW of P1501 shall be greater than 1.000 ohm.

3.3.5 Life. Life requirements for unit conditions (see 6.3.1) shall be as follows:

3.3.5.1 Operating. The power unit shall meet requirements of 3.3.1 through 3.3.3.5 operation throughout its mission time of nine minutes.

3.3.5.2 Total operating life. The power unit shall be capable of satisfactory operation for a minimum of 100 operating hours total from date of acceptance by the procuring activity, and thereafter shall meet the performance requirements and product characteristics of this specification (Column II values).

3.4 Environmental requirements. Unless otherwise specified the power unit shall be capable of conforming to performance requirements and product characteristics requirements of this specification (Column IB values), after being exposed, at any temperature in the applicable range, to the environments under all conditions specified in this section.

3.4.1 Vibration. The power unit shall be capable of withstanding vibration shown in TABLE II.

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TABLE II

Vibration schedule.

Condition (see 6.3.1)	Double amplitude (in) minimum	Acceleration (g's) minimum	Frequency (Hz)
Ready Nonoperating and Ready	0.2	- - -	5 to 17
	- - -	3.0	17 to 500

3.4.2 Shock. The power unit shall be capable of withstanding the shock specified in TABLE III.

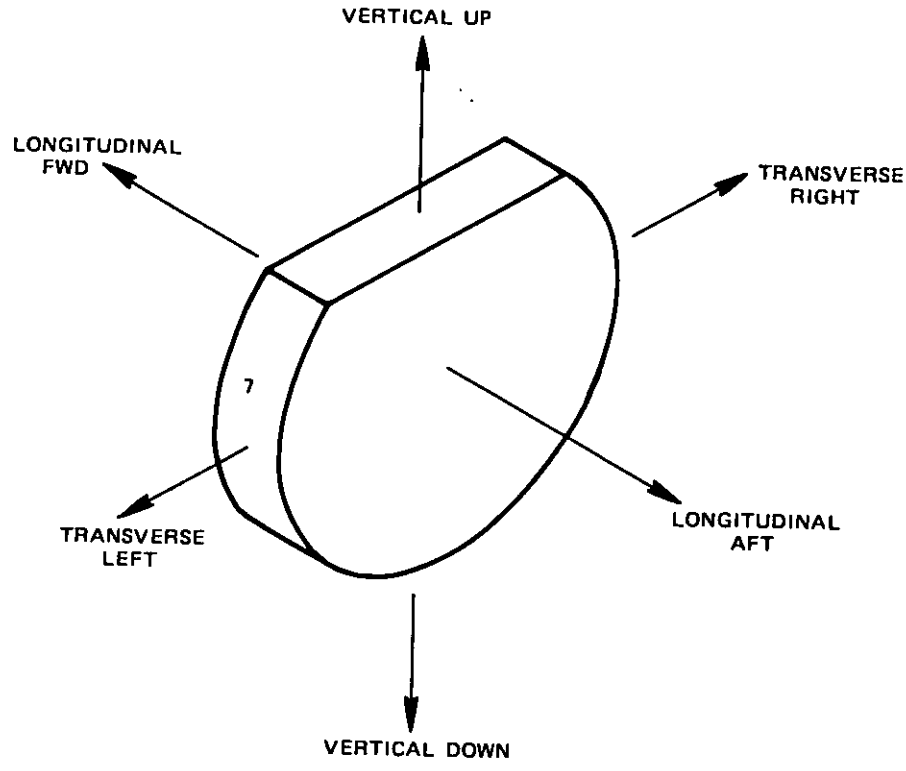
TABLE III

Shock schedule.

Condition (see 6.3.1)	Peak magnitude (g's)	Duration (milliseconds)	Acceleration direction*	Number of Impacts
Ready nonoperating	60 ±10%	8.0 to 10.0	3 fwd or aft axis 3 left or right axis 3 up or down axis	9
Ready	45 ±10%	18.0 to 20.0	Long, fwd axis	3
	85 ±10%	0.5 to 1.5	3 up or down axis 3 right or left axis	6
	300 ±10%	0.5 to 1.5	Long. aft axis	3

*See FIGURE 4 for axes of power unit.

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Note: The left side is the side on which the module number appears. Type II power units consist of two modules, which are designated (forward to aft) as A6A1 and A6A2.

FIGURE 4. Axes of the power unit

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3.4.3 Acceleration. The power unit shall be capable of withstanding accelerations specified in TABLE IV.

TABLE IV

Acceleration schedule

Condition (see 6.3.1)	Magnitude (g's)	Acceleration Direction
Ready nonoperating	25 ±10%	Longitudinal Forward
Ready	35 ±10%	Longitudinal Aft

3.4.4 Temperature. The power unit shall be capable of withstanding temperatures for the following specified conditions:

3.4.4.1 Nonoperating and ready nonoperating. The power unit shall be capable of withstanding ambient temperatures between -45°F and 140°F wherein the duration of the upper temperature does not exceed four hours at any one time, and the temperature is stabilized in a temperature range of 60°F to 95°F prior to any performance test.

3.4.4.2 Ready and operating. The power unit, when operated in an environment with a temperature between 32°F and 98°F, shall meet the performance and product characteristics of this specification (Column II values) after exposure to temperatures between 0°F to 140°F, for a period wherein the duration of the upper temperature limit does not exceed four hours at any one time.

3.4.5 Pressure. The power unit shall be capable of withstanding pressures of from 2.7 to 20 psia.

3.4.6 Humidity. The power unit shall withstand exposure to 95 percent relative humidity at +110 degrees F for eight hours without degradation of finish or functional capabilities.

3.5 Workmanship. Workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with requirements of this specification. All parts used in the power unit shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classifications of inspection. Inspections of the power unit shall be classified as follows:

a. First article inspection

1. First article sample (see 4.3.1)

2. First article tests (see 4.5)

b. Periodic production inspection

1. Periodic production sampling (see 4.3.2)

2. Periodic production tests (see 4.6)

c. Quality conformance inspection

1. Quality conformation sampling (see 4.3.3)

2. Quality conformance tests and examinations (see 4.7)

4.3 Sampling.

4.3.1 First article sample. A sample of three power units shall be subjected to tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the sample shall be based on no defects in the sample.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed the quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

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a. Step 1. Upon initiation of production, randomly select one unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively, starting with Sample Number 1.

b. Step 2. Randomly select one unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively, starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective unit. When a defective unit (see 6.3.8) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand, and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

e. Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. Tests and examinations specified in 4.7 shall be conducted on each power unit to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the power unit shall be subjected to tests under following conditions:

a.	Temperature	Room ambient (60 to 85 degrees F)
b.	Humidity (relative)	Room ambient, up to 95 percent
c.	Altitude	Normal ground atmospheric pressure
d.	Vibrations	None

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4.5 First article tests. The power unit within the first article sample (identified as power units 1, 2, and 3) shall be subjected to tests specified in TABLE V. Certified inspection data covering results of first article tests shall be forwarded to the Naval Sea Systems Command. Certified inspection data shall consist of the following:

a. Date of tests and names of persons conducting tests

b. Run or test plan used

c. Serial number of the power unit

d. Test results of first article tests (see 4.5.2 through 4.5.3).

TABLE V

First article test schedule.

Test Sequence	Test description	Power unit number		
		1	2	3
1	Quality conformance tests and examinations (4.7) with Column IA values specified in 3.3	X	X	X
2	Vibration (4.5.2.1)	X		X
3	Shock (4.5.2.2)	X	X	
4	Acceleration (4.5.2.3)	X		X
5	Temperature, storage (4.5.2.4.1)	X	X	
6	Temperature, operating (4.5.2.4.2)	X	X	
7	Pressure test (4.5.2.5)	X	X	
8	Humidity test (4.5.2.6)	X		X
9	Quality conformance tests and examinations (4.7) with Column IB values specified in 3.3	X	X	X
10	Life test (4.5.2.7)	X		
11	Quality conformance tests and examinations (4.7) with Column II values specified in 3.3	X		

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4.5.1 First article test equipment. The following items of test equipment are required to perform first article tests specified herein:

4.5.1.1 A vibration table capable of vibrating the power unit, plus fixture weight, at 3 g's maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitudes of 0.2 inch.

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 20 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g's magnitude for a period of one minute.

4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of -45 to +140 degrees F.

4.5.1.5 A humidity chamber capable of providing a constant humidity condition of 95 percent at 110 degrees F for extended periods up to 8 hours.

4.5.1.6 Items of test equipment specified in 4.7.

4.5.2 Detailed first article tests.

4.5.2.1 Vibration.

4.5.2.1.1 Ready nonoperating and ready. The power unit, mounted on a support frame, shall be rigidly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in TABLE II to the Power Unit along the axes shown in FIGURE 4. Vibration will be conducted at three temporature conditions: the power unit stabilized at room temperature, at -45 +0 -5°F and +140 +5 -0°F. The total vibration time in each axis shall be 180 minutes with 60 minutes of this time with power unit stabilized at -45 +0 -5°F; 60 minutes with power unit stabilized at +140 +5 -0°F; and 60 minutes with power unit stabilized at room temperature. The frequency of applied vibration shall be cycled between the frequency limits and at the vibratory levels of TABLE II. TABLE II shall be considered as input to the test item. The frequency rate during cycling shall be logarithmic such that 1/12 of the total vibration time at each axis is required to proceed from the low to higher and back to the lower extremity of the frequency range specified in TABLE II.

4.5.2.2 Shock.

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4.5.2.2.1 Ready, nonoperating and ready. Mount the power unit onto a shock test fixture so that a minimum of 90 percent of table input of the fixture will be applied to the power unit under test. Subject the power unit to shocks listed in TABLE III, in the conditions specified. Duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock.

4.5.2.3 Acceleration. The unpackaged power unit shall be subjected to acceleration magnitudes and directions required by TABLE IV for one minute at room temperature for each condition specified.

4.5.2.4 Temperature.

4.5.2.4.1 Temperature, ready nonoperating. The power unit shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

a. Heat chamber to +140 +5 -0 degrees F in 2.5 ±0.5 hours.

b. Maintain +140 +5 -0 degrees F for 4 hours minimum, then within 2.5 ±0.5 hours cool chamber to -45 +0 -5 degrees F.

c. Maintain at -45 +0 -5 degrees F for 4 hours minimum, then within 2.5 ±0.5 hours return chamber to room temperature.

Repeat this schedule for a minimum of three cycles.

4.5.2.4.2 Temperature, operating. The power unit, mounted in a manner thermally representative of its final mounting in the torpedo, shall meet the quality conformance test requirements of 4.7 except as specified below:

a. The power unit shall be stabilized for one hour at a temperature of +140 +5 -0 degrees F, then transferred to an environment at +98 +0 -5 degrees F and tested to ensure conformance with performance requirements and product characteristics of this specification (Column II values). This conformance shall be shown to exist for any combination of power supply and input voltages tabulated in 3.3.1.1 (Column II) and 3.3.1.2 (Column II) of this specification.

b. The power unit shall be stabilized for one hour at a temperature of zero +0 -5 degrees F, then transferred to an environment at +32 +3 -0 degrees F and tested to ensure conformance with performance requirements and product characteristics of this specification (Column II values). This conformance shall be shown to exist for any combination of power supply and input voltages tabulated in 3.3.1.1 (Column II) and 3.3.1.2 (Column II) of this specification.

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4.5.2.5 Pressure. Subject the power unit to pressures from 2.7 to 20.0 psia.

4.5.2.6 Humidity. The power unit shall be placed in a suitable test chamber and subjected to a relative humidity of at least 95 percent (but in no case shall moisture be allowed to drip on the unit) at a temperature of 110°F +5°F -0°F for a period of 8 hours minimum. Upon completion of this test, remove the power unit from the chamber and, after exposure to normal test conditions (4.4) for a period of 4 to 8 hours, initiate the quality conformance tests (4.7).

4.5.2.7 Life test. The power unit shall be energized as described below for a total of 200 duty cycles. Each duty cycle shall consist of 10.0 -0.0 +0.5 minutes energized and 20.0 -0.0 +1.0 minutes de-energized. To apply towards the completion of the required 200 duty cycles, a minimum of 10 consecutive duty cycles shall be accomplished during each test period. During the life test, a maximum of 90 hours will be allowed between the end of one duty cycle and the start of the next duty cycle.

4.5.2.7.1 Test set modification. The following temporary modifications shall be made to the power unit test set to be used for the life test. (The test set modified as defined below can still be used for normal quality conformance testing if item (3) is mechanized with a switch and the cards in item (4) are replaced.)

a. Actuator controller test panel assembly

1. Connect a 3.9K 1/2 W resistor from INSTRUMENT 1 (TP15) to the UP/DOWN switch (S3) DOWN terminal.

2. Connect a 3.9K 1/2 W resistor from INSTRUMENT 1 (TP15) to the LEFT/RIGHT switch (S2) RIGHT terminal.

3. Disconnect R11, R13 and R15.

4. Remove plug-in circuit boards TB2, TB3 and TB4.

5. Connect LEFT UPPER RUDDER (TP1) to RIGHT UPPER RUDDER (TP2).

6. Connect LEFT LOWER RUDDER (TP3) to RIGHT LOWER RUDDER (TP4).

7. Connect UP ELEVATOR (TP5) to DOWN ELEVATOR (TP6).

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b. Power supply test panel

1. Change the SHUNT FIELD load switch (S103) to a 3 position on-off momentary on switch to allow either momentary or continuous connection of the field load resistor (R121, 20 ohms, 180 watt).

2. Install an ON-OFF switch in the DC output of the +40V and -40V power supplies.

c. Proceed as in 4.7.3.1.2 and 4.7.3.1.3.

4.5.2.7.2 Initial test setup.

a. Set controls on actuator controller test panel as follows:

Control	Setting
CW/CCW	CW (for first 100 30-minute duty cycles)
CW/CCW	CCW (for second 100 30-minute duty cycles)
NORMAL/PULSE	NORMAL
LEFT/RIGHT	RIGHT
UP/DOWN	DOWN

b. Connect the 600 ohm output of an HP211A, or similar, square wave generator to INSTRUMENT 1 and test set common. Adjust the output of the generator to 5 cps, adjust the symmetry to give 50% duty cycle and the output voltage to -20V peak.

c. Set controls on power supply test panel as follows:

Control	Setting
LOAD	MAX (for first 100 30-minute duty cycles)
LOAD	MIN (for second 100-30 minute duty cycles)
COMPUTER ENABLE	ON
EXPLODER ARM	OUT
SHUNT FIELD	IN

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4.5.2.7.3 Test procedure.

a. Turn test set MAIN POWER to ON position.

CAUTION

The cooling ring blower shall be operating at all times during the life test.

b. With the power switches of the ACTUATOR CONTROLLER PANEL and POWER SUPPLY TEST PANEL in the ON position, monitor the appropriate test points on the POWER SUPPLY TEST PANEL and adjust the +40V supply to +40.0 ±0.5 vdc.

NOTE: In (c) and (d) the SHUNT FIELD voltage should be monitored periodically. If the indication is outside the 10.0 ±2.0 volt magnitude limit, readjust the SHUNT FIELD voltage magnitude to 10.0 ±0.5 vdc and continue the test.

c. During the first 100 30-minute duty cycles, decrease the -40V supply toward -39.5 vdc (with FIELD LOAD switch in the IN position) until the SHUNT FIELD voltage is +10.0 ±0.5 vdc. The SHUNT FIELD voltage shall be maintained at +10.0 ±2.0 vdc.

d. During the second 100 30-minute duty cycles, increase the -40V supply toward -41.5 vdc (with FIELD LOAD switch in the IN position) until the SHUNT FIELD voltage is -10.0 ±0.5 vdc. Maintain the SHUNT FIELD voltage at -10.0 ±2.0 vdc.

e. Monitor the two +90 VDC outputs (located on the power panel) for indication of power unit operation. Normal operation is indicated by a +90 to +100 vdc reading.

f. Each duty cycle shall be accomplished by turning the +40V and -40V power supplies ON and OFF.

g. After completion of the 200 30-minute duty cycles, remove the square wave generator from the INSTRUMENT 1 test jack on the ACTUATOR CONTROLLER test panel.

h. Set the SHUNT FIELD switch to OUT position (20 ohm, 180 WATT load disconnected).

4.5.3 First article acceptance criteria. Acceptance shall be based on satisfactory completion of test sequence item 9 or 11 in TABLE V as applicable.

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4.6 Periodic production tests. Periodic production samples, selected in accordance with 4.3.2, shall be subjected to the tests of 4.5.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for all tests of Sections 4.5 and 4.7.

4.6.2 Periodic production test schedule. The periodic production test sample shall be subjected to the tests noted in TABLE VI in the order specified.

TABLE VI

Periodic production test schedule.

Test	Sample and test sequence		
	Sample No. 1, 4, etc.	Sample No. 2, 5, etc.	Sample No. 3, 6, etc.
Quality conformance (Column IA) (4.7)	1	1	1
Vibration (4.5.2.1)	2*		2*
Shock (4.5.2.2)		2*	5*
Acceleration (4.5.2.3)			4*
Temperature nonoperating (4.5.2.4.1)		3	6
Hi-temperature operating (4.5.2.4.2(a))		4	7
Lo-temperature operating (4.5.2.4.2(b))	3		3
Pressure (4.5.2.5)			8
Humidity (4.5.2.6)			9
Quality conformance (Column IB) (4.7)	4	5	10
Life (4.5.2.7)			11
Quality conformance (Column II) (4.7)			12

* Indicates visual examination after test.

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4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2.1 and 4.7.1.3.

4.7 Quality conformance tests and examinations. Unless otherwise specified, each power unit delivered under contract shall be be subjected to the tests and examinations specified herein. In addition, each power unit delivered under contract as a spare FIR unit shall be assembled into a forebody in accordance with Dwg 2539311 Nose Section and DWG 2539026 Control Group and subjected to applicable tests and examinations specified in the quality conformance paragraph of MIL-T-82426 Main Assembly. The forebody configuration shall consist of the above spare FIR item in combination with four other spare FIR items or combination of FIR spare and "slave" units.

4.7.1 Conditions.

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in the order shown.

4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in Reading column indicating that requirements were satisfactorily met. A copy of the report form, bearing recorded data, shall be shipped with the inspected unit.

4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units which have not been corrected will be considered for acceptance only upon formal application by the contractor to the Naval Sea Systems Command via the Government representative.

4.7.2 Quality conformance test equipment. Items of test equipment shown in TABLE VII are required to perform the quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the

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cognizant Government inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

4.7.3 Electrical tests.

4.7.3.1 Initial test setup.

4.7.3.1.1 Perform the following tests on power unit prior to making any test set connections:

a. Connect multimeter between pin UU (TORPEDO COMMON) of connector P1501 and each of the following pins of connector P1501:

F (RECEIVER COMMON)

H (RECEIVER COMMON)

J (RECEIVER COMMON)

U (COMMON)

Y (400 Hz GROUND)

SS (FIELD RETURN)

VV (TORPEDO COMMON)

b. Multimeter shall indicate 0.5 ohm or less for any combination of above connections.

c. Connect multimeter between power unit casting and the following pins of connector P1501:

A (EXPLODER ARM)

B (+90 VDC)

R (+30 VDC)

S (-30 VDC)

T (+5 VDC)

HH (-40 VDC STEADY STATE)

TT (+40 VDC STEADY STATE)

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UU (TORPEDO COMMON)

WW (+40 VDC STEADY STATE)

XX (-40 VDC STEADY STATE)

d. Multimeter shall indicate a resistance greater than 500,000 ohms between casting and pins.

e. Connect multimeter between pins HH (-40 VDC STEADY STATE) and XX (-40 VDC STEADY STATE) of connector P1501. Multimeter shall indicate a resistance greater than 500 ohms

f. Connect multimeter between pins TT (+40 VDC STEADY STATE) and WW (+40 VDC STEADY STATE) of connector P1501. Multimeter shall indicate a resistance greater than 1000 ohms.

4.7.3.1.2 Position all power unit test set POWER switches to OFF. Mount power unit in cooling ring and connect J1501 of power unit test set cable to P1501 of power unit.

4.7.3.1.3 Position POWER switches on -40 volt regulated power supply and +40 volt regulated power supply to ON. Position power panel circuit breaker switch to ON. Position RUNNING TIME meter to zero minutes.

4.7.3.2 +36 vdc and -36 vdc Input (maximum load).

4.7.3.2.1 Position power supply test panel switches as follows:

Switch	Position
COMPUTER ENABLE	OFF
LOAD	MAX
EXPLODER ARM	OUT
METER SWITCH	OFF

NOTE: SHUNT FIELD switch is a momentary contact switch, spring loaded to OUT position.

4.7.3.2.2 Position power supply test panel POWER switch to ON.

CAUTION

When using digital voltmeter, ensure that it is not grounded.

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TABLE VII

Quality conformance test equipment.

Item	Identification
Power unit test set	Dwg 2450730
Digital voltmeter (Range: 0.001 to 999.9 vdc or vrms; DC accuracy: ±0.05% of reading ±1 digit; AC accuracy: ±0.1% of reading ±2 counts; Input impedance; 10 megohms, 20 pf.)	Hewlett-Packard, Model 3440A with 3445A AC/DC Range Unit
Electronic counter (Frequency range: 10Hz to 120 kHz; Accuracy ±1 count ± time base accuracy.)	Hewlett-Packard Model 522B
Distortion analyzer (Frequency range: 20 Hz to 20 kHz; Accuracy: ±3% of full scale for distortion levels as low as 0.5%)	Hewlett-Packard Model 330B
DC VTVM (Range: 1 millivolt full scale to 1000 volts full scale; Accuracy: ±1% full scale.)	Hewlett-Packard Model 412A
Oscilloscope (Bandpass: dc to 15 mc; Rise Time: 0.023 usec; Accuracy: -0.05 to 20 v/cm, 3%.)	Tektronix Model 535A with Type C-A Preamplifier

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TABLE VII (Con't)

Quality conformance test equipment.

Item	Identification
10X Passive Probe (Input resistance: 10 megohms, Input capacitance: 7 pf with 20 pf input, 9.5 pf with 47 pf input; Voltage rating: 60 dc or ac (peak-to-peak); Attenuation: 10X)	Tektronix Model P6006
Multimeter (Resistance 0 to 20 megohms.)	Simpson Model 260
Capacitor (1000 pf, 10%, 500 volt)	CM20BD102KN3

4.7.3.2.3 Connect digital voltmeter to power supply test panel -40V jack and adjust voltage controls on -40 volt regulated power supply to -36.0 ±0.1 vdc. The -40 volt regulated power supply ammeter shall indicate between 1.5 and 5.0 amperes dc.

4.7.3.2.4 Connect digital voltmeter to power supply test panel +40V jack and adjust voltage control on +40 volt regulated power supply to +36.0 ±0.1 vdc. The +40 volt regulated power supply ammeter shall indicate between 0.5 and 3.0 amperes dc.

4.7.3.2.5 Connect digital voltmeter to following power supply test panel jacks. Measured voltage shall be within limits specified.

Test point	Limits (vdc)
+5V	+4.75 to +5.25
-15V	-14.5 to -15.5
-30V	-29.50 to -31.2
-30VG	-28.9 to -30.9
+90V (upper)	+90.0 to +100.0
+15V	+14.5 to +15.5
+30V	+29.4 to +30.6
+90V (lower)	+90.0 to +100.0

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4.7.3.3 4.8 KC supply (maximum load).

4.7.3.3.1 Connect digital voltmeter to 4.8 KC IN Ø jack. Voltage shall be between 25.5 and 26.5 volts rms.

4.7.3.3.2 Connect electronic counter to 4.8 KC IN Ø jack. Frequency shall be between 4770 and 4830 Hz.

4.7.3.3.3 Position distortion analyzer controls as follows:

Control	Position
INPUT (AF/RF)	AF
FUNCTION	SET LEVEL
METER RANGE	100%
FREQUENCY RANGE	X100
FREQUENCY	Approximately 48

4.7.3.3.4 Connect distortion analyzer AF INPUT terminals to 4.8 KC IN Ø jack. Adjust INPUT SENSITIVITY control until meter pointer reaches full scale deflection of 1.0. Position FUNCTION switch to DISTORTION. Adjust FREQUENCY and BALANCE controls and position METER RANGE switch for minimum meter indication. Indicated distortion shall not exceed 3 percent.

4.7.3.4 +45 and -46 vdc input (minimum load).

4.7.3.4.1 Position LOAD switch to MIN. Connect the digital voltmeter to power supply test panel jack +40V, and adjust voltage control on the +40 volt regulated power supply to +46.0 ±0.1 vdc.

4.7.3.4.2 Connect probe from channel A of oscilloscope to blue collector lead terminal of Q1508 (located on TB1501) and oscilloscope ground to common ground. Set channel A to 0.5 V/CM and DC. Adjust the trace to bottom line of graticule. (Paragraph 4.7.3.4.2 is not required on Power Unit DL 2557500.)

4.7.3.4.3 Turn off the plus and minus power supplies for a period of at least 5 minutes, and then turn power supplies on. (Paragraph 4.7.3.4.3 is not required on Power Unit DL 2557500.)

4.7.3.4.4 While observing the trace on the oscilloscope, touch a 1000 pf capacitor successively from the casting to the blue collector lead terminal, the green base lead terminal, and the yellow emitter lead

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terminal of Q1508. There shall be no sustained oscillations indicated on the oscilliscope for longer than 10 msec after removal of the capacitor from each respective lead, except during the first 1/2 second after power turn-on. (Paragraph 4.7.3.4.4 is not required on Power Unit DL 2557500.)

4.7.3.4.5 Connect the digital voltmeter to power supply test panel jack -40V and adjust voltage control on -40 volt regulated power supply to -46.0 ±0.1 vdc.

4.7.3.4.6 Connect digital voltmeter to the following power supply test panel jacks. Measured voltage shall be within limits specified.

Test Point	Limits (vdc)
+5V	+4.75 to +5.25
-15V	-14.5 to -15.5
-30V	-29.50 to -30.80
-30VG	-28.9 to -30.5
+90V (upper)	+90.0 to +100.0
+15V	+14.5 to +15.5
+30V	+29.4 to +30.6
+90V (lower)	+90.0 to +100.0

4.7.3.5 4.8 KC supply (minimum load).

4.7.3.5.1 Connect digital voltmeter to 4.8 KC IN Ø jack. Voltage shall be between 25.5 and 26.5 volts rms.

4.7.3.5.2 Connect electronic counter to 4.8 KC IN Ø jack. Frequency shall be between 4770 and 4830 Hz.

4.7.3.5.3 Position distortion analyzer controls as follows:

Control	Position
INPUT (AF/RF)	AF
FUNCTION	SET LEVEL
METER RANGE	100%
FREQUENCY RANGE	X100
FREQUENCY	Approximately 48

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4.7.3.5.4 Connect distortion analyzer AF INPUT terminals to 4.8 KC IN Ø jack. Adjust INPUT SENSITIVITY control until meter pointer reaches full scale deflection of 1.0. Position FUNCTION switch to DISTORTION. Adjust FREQUENCY and BALANCE controls and position METER RANGE switch for minimum meter indication. Indicated distortion shall not exceed 3 percent.

4.7.3.6 Exploder arm.

4.7.3.6.1 Connect digital voltmeter to power supply test panel -40V jack and adjust voltage control on -40 volt regulated power supply to -40.0 ±0.1 vdc.

4.7.3.6.2 Connect digital voltmeter to power supply test panel +40V jack and adjust voltage control on +40 volt regulated power supply to +40.0 ±0.1 vdc.

4.7.3.6.3 Connect digital voltmeter minus lead to EXP-ARM jack (voltmeter positive lead connect to +40V jack). Position EXPLODER ARM switch to IN. Voltage shall be -0.1 to +0.4 vdc maximum.

4.7.3.6.4 Connect positive lead of digital voltmeter to -40V jack. Position EXPLODER ARM switch to OUT and COMPUTER ENABLE switch to ON. Digital voltmeter indication shall be 0.0 to -1.5 vdc. Digital voltmeter indications shall not vary more than 0.5 vdc when observed for at least 30 seconds.

4.7.3.6.5 Position EXPLODER ARM switch to IN. Digital voltmeter indication shall be between 0.0 and -1.5 vdc.

4.7.3.6.6 Position EXPLODER ARM switch to OUT and COMPUTER ENABLE switch to OFF. Connect negative lead of digital voltmeter to GND jack.

4.7.3.7 Voltage regulator.

NOTE: SHUNT FIELD switch (momentary) should only be in IN position while reading indicated voltages on DC VTVM or adjusting supply voltage.

4.7.3.7.1 Connect digital voltmeter to power supply test panel -40V jack. Position SHUNT FIELD switch to IN and adjust voltage control on -40 volt regulated power supply to -37.5 ±0.5 vdc.

4.7.3.7.2 Connect digital voltmeter to power supply test panel +40V jack. Position SHUNT FIELD switch to IN and adjust voltage control on +40 volt regulated power supply to +45.0 ±0.1 vdc.

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4.7.3.7.3 Connect dc vtvm positive lead to SHUNT FIELD (+) and negative lead to SHUNT FIELD (-). Operate for about 15 minutes or until thermal equilibrium is achieved, Position SHUNT FIELD switch to IN. The dc vtvm shall indicate between 0.0 and +1.5 vdc. If dc vtvm indicates more than 1.5v, increase +40 vdc regulated supply from initial setting until SHUNT FIELD voltage reads less than 1.5 volts. Do not exceed +45.5 vdc during adjustment. Maximum allowable voltage on +40 vdc is 45.2 volts. For A, 1B, and Column II limits, the maximum voltage on +40 vdc remains at 4' 2 volts with the same adjustment procedure for determining the allowabl. SHUNT FIELD voltage.

4.7.3.7.4 Position SHUNT FIELD switch to IN and adjust +40 volt regulated power supply to +40.0 ±0.5 vdc. Position SHUNT FIELD switch to IN. DC VTVM shall indicate between +30.0 and +40.0 vdc.

4.7.3.7.5 Connect digital voltmeter to power supply test panel -40V jack. Position SHUNT FIELD switch to IN. Adjust voltage control on -40 volt regulated power supply until indication on DC VTVM is 0.0 ±0.2 vdc. Digital voltmeter shall indicate between -40.0 and -41.2 vdc.

4.7.3.7.6 Position SHUNT FIELD switch to IN and adjust voltage control on -40 volt regulated power supply to -42.5 ±0.5 vdc. Position SHUNT FIELD switch to IN. DC VTVM shall indicate between -6.0 and -42.0 vdc.

4.7.3.7.7 Position power supply test panel power switch to OFF. Record indicated RUNNING TIME. Disconnect test equipment leads from power supply test panel.

4.7.3.8 Actuator controller.

4.7.3.8.1 Position actuator controller test panel switches as follows:

Switch	Position
NORMAL/PULSE	NORMAL
ELEVATOR PITCH	UP
RUDDERS YAW	RIGHT
RUDDERS ROLL	CW

4.7.3.8.2 Record time indicated on RUNNING TIME meter. Position actuator controller test panel POWER switch to ON. Power supply test panel POWER switch shall remain OFF. The CONTINUITY lamp shall light.

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4.7.3.8.3 Connect digital voltmeter to actuator controller test panel +4CV jack and adjust voltage control on +40 volt regulated power supply to +40.0 ±0.5 vdc.

4.7.3.8.4 Connect digital voltmeter to actuator controller test panel -40V jack and adjust voltage control on -40 volt regulated power supply to -40.0 ±0.5 vdc.

4.7.3.8.5 Monitor +40 vdc and -40 vdc current on actuator controller test panel ammeters. The +40 volt current shall be 0.85 to 1.55 amperes. The -40 volt current shall be 1.65 to 1.95 amperes.

4.7.3.8.6 Observe ELEVATOR and RUDDERS lamps. The ELEVATOR UP and RUDDERS UPPER and LOWER RIGHT CW lamps shall be lighted. Remaining ELEVATOR and RUDDERS lamps shall be extinguished.

4.7.3.8.7 Connect positive lead of digital voltmeter to FUNCTION MONITOR jack and negative lead to COMMON jack. Rotate FUNCTION switch to positions 1 through 6. Depress PUSH-TO-TEST switch when observing FUNCTION voltages on digital voltmeter. Verify that following voltages are indicated on digital voltmeter.

CAUTION

The PUSH-TO-TEST switch must not be depressed while rotating FUNCTION switch.

FUNCTION switch position	Voltage reading
1 - (LEFT UPPER RUDDER)	+0.20 to -0.10
2 - (RIGHT UPPER RUDDER)	-21.8 to -25.8 vdc
3 - (LEFT LOWER RUDDER)	+0.10 to -0.20 vdc
4 - (RIGHT LOWER RUDDER)	+34.0 to +37.0 vdc
5 - (UP ELEVATOR)	-34.0 to -37.0 vdc
6 - (DOWN ELEVATOR)	+0.20 to -0.10 vdc

4.7.3.8.8 Position RUDDERS ROLL switch to CCW. The RUDDER UPPER and LOWER RIGHT CW lamps shall extinguish and RUDDERS UPPER and LOWER RIGHT CCW lamps shall light.

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4.7.3.8.9 Monitor +40 vdc and -40 vdc current on actuator controller test panel ammeters. The +40 volt current shall be 0.65 to 1.35 amperes. The -40 volt current shall be 1.95 to 2.25 amperes.

4.7.3.8.10 Rotate FUNCTION switch to positions 2 and 4. Verify that following voltages are indicated on digital dc voltmeter when PUSH-TO-TEST switch is depressed:

CAUTION

The PUSH-TO-TEST switch must not be depressed while rotating FUNCTION switch.

FUNCTION switch position	Voltage reading
2	-34.0 to -37.0 vdc
4	+21.8 to +25.8 vdc

4.7.3.8.11 Position ELEVATOR PITCH switch to DOWN, RUDDERS YAW switch to LEFT, and RUDDERS ROLL switch to CCW. The ELEVATOR DOWN and RUDDERS UPPER and LOWER LEFT CCW lamps shall light. The remaining ELEVATORS and RUDDERS lamps shall be extinguished.

4.7.3.8.12 Monitor +40 vdc and -40 vdc current on actuator controller test panel ammeters. The +40 volt current shall be 1.40 to 2.10 amperes. The -40 volt current shall be 1.25 to 1.55 amperes.

4.7.3.8.13 Rotate FUNCTION switch to position 1 through 6. Verify that following voltages are indicated on digital voltmeter when PUSH-TO-TEST switch is depressed:

CAUTION

The PUSH-TO-TEST switch must not be depressed while rotating FUNCTION switch.

FUNCTION switch position	Voltage reading
1	+21.8 to +25.8 vdc
2	+0.10 to -0.20 vdc
3	-34.0 to -37.0 vdc
4	+0.20 to -0.10 vdc
5	+0.10 to -0.20 vdc
6	+34.0 to +37.0 vdc

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4.7.3.8.14 Position RUDDERS ROLL switch to CW. The RUDDERS UPPER and LOWER LEFT CCW lamps shall extinguish, and RUDDERS UPPER and LOWER LEFT CW lamps shall light.

4.7.3.8.15 Monitor +40 vdc and -40 vdc current on actuator controller test panel ammeters. The +40 volt current shall be 1.65 tO 2.35 amperes. The -40 volt current shall be 0.95 to 1.25 amperes.

4.7.3.8.16 Rotate FUNCTION switch to positions 1 and 3. Verify that following voltages are indicated on digital voltmeter when PUSH-TO-TEST switch is depressed.

CAUTION

The PUSH-TO-TEST switch must not be depressed while rotating FUNCTION switch.

FUNCTION switch position	Voltage reading
1	+34.0 to +37.0 vdc
3	-21.8 50 -25.8 vdc

4.7.3.9 Controller oscillator.

4.7.3.9.1 Position oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC (LF REJECT)
TRIGGER SLOPE	INT +
STABILITY	FULL CW
TRIGGER LEVEL	FULL CCW
HORIZONTAL DISPLAY	A
TIME/CM	0.5 ms
VARIABLE (TIME/CM)	CALIBRATED
VOLTS/CM	1
POLARITY	NORMAL
MODE	A ONLY
AC/DC	DC

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4.7.3.9.2 Connect oscilloscope channel A using X10 passive probe to POWER UNIT OSC test jack. Connect oscilloscope ground to actuator controller test panel COMMON jack. Center oscilloscope trace on graticule. Using oscilloscope HOTIZONTAL position control, adjust start of trace to coincide with far left vertical line of graticule. Adjust stability control for a stable trace. Waveform voltages and periods, as indicated in FIGURE 2, shall be as follows:

Reading	Indication
V1	-1.1 vdc max
V2	-22.5 to -24.5 vdc
T1	3.6 to 5.2 cm

4.7.3.9.3 Set the TIME/CM switch to 0.2 ms or 0.1 ms as required and adjust the VARIABLE (TIME/CM) control such that period T1, as indicated in FIGURE 2, covers 10.0 cm of oscilloscope graticule. Period T2 shall be 6.4 to 7.0 cm.

4.7.3.9.4 Position actuator controller test panel POWER switch to OFF. Disconnect oscilloscope.

4.7.3.10 Pulse current.

4.7.3.10.1 Position oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC (LF REJECT)
TRIGGER SLOPE	INT +
STABILITY	FULL CW
TRIGGER LEVEL	FULL CCW
HORIZONTAL DISPLAY	A
TIME/CM	0.2 ms
VARIABLE (TIME/CM)	CALIBRATED
VOLTS/CM	1 (if X10 probe is used)
POLARITY	NORMAL
MODE	A ONLY
AC/DC	DC

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4.7.3.10.2 Adjust oscilloscope trace to horizontal center line. Connect oscilloscope channel A to actuator controller test panel FUNCTION MONITOR jack. Position FUNCTION switch to 1, and NORMAL/PULSE switch to PULSE.

CAUTION

The PUSH-TO-TEST switch must not be depressed while rotating FUNCTION switch.

4.7.3.10.3 Position actuator controller test panel POWER switch to ON. Monitor oscilloscope waveform at FUNCTION MONITOR jack for FUNCTION switch positions 1, 4, and 6. While monitoring oscilloscope waveform, depress PUSH-TO-TEST switch. Adjust oscilloscope triggering level and stability, as required to obtain a stable trace. Pulse amplitude of voltage V, as shown in FIGURE 2, shall be +16 to +25 vdc at each switch position.

4.7.3.10.4 Position oscilloscope TRIGGER SLOPE control to INT -. Monitor oscilloscope waveform at FUNCTION MONITOR jack for FUNCTION switch positions 2, 3, and 5. Monitor oscilloscope waveforms as described in 4.7.3.10.3 above. Pulse amplitude of voltage, as shown in FIGURE 2, shall be -16 to -25 vdc at each switch position.

4.7.3.10.5 Position oscilloscope controls as follows:

Control	Position
TRIGGER SLOPE	as indicated
STABILITY	FULL CW
TRIGGER LEVEL	FULL CCW
TIME/CM	as indicated
VOLTS/CM	1 volt/variable

4.7.3.10.6 Connect oscilloscope trigger input to actuator controller test panel RUDDERS YAW jack when monitoring oscilloscope waveforms at FUNCTION MONITOR jack for switch positions 1 through 4. Connect oscilloscope trigger input to actuator controller test panel ELEVATOR PITCH jack when monitoring oscilloscope waveforms at FUNCTION MONITOR jack for switch positions 5 and 6. Position FUNCTION switch to position 1. Depress PUSH-TO-TEST switch while adjusting oscilloscope TRIGGERING controls to obtain a stable trace. Refer to 4.7.3.10.3 and using oscilloscope variable VOLTS/CM controls, adjust oscilloscope waveform peak-to-peak

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amplitude to cover 2.0 cm of oscilloscope graticule. The fall time T_f is measured from 90% point to 10% point of waveform, as shown in FIGURE 5. Repeat above procedure for FUNCTION switch positions 2 through 6. Oscilloscope settings and indicated fall times are as follows:

FUNCTION	Switch position	TIME/CM	TRIGGER SLOPE	Fall time
1		5 usec	EXT-	25 usec max
2		10 usec	EXT+	50 usec max
3		10 usec	EXT-	50 usec max
4		5 usec	EXT+	25 usec max
5		10 usec	EXT-	50 usec max
6		5 usec	EXT+	25 usec max

4.7.3.10.7 Position actuator controller test panel POWER switch to OFF. Disconnect oscilloscope. Set NORMAL/PULSE switch to NORMAL.

4.7.3.11 Pitch inhibit.

4.7.3.11.1 Connect oscilloscope channel A to PITCH INHIBIT jack. Position oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC (LF REJECT)
TRIGGER SLOPE	INT-
STABILITY	FULL CW
TIME/CM	1 MS
VARIABLE (TIME/CM)	CALIBRATED
VOLTS/CM (CHANNEL A)	1
POLARITY	NORMAL
MODE	A ONLY
AC/DC	DC

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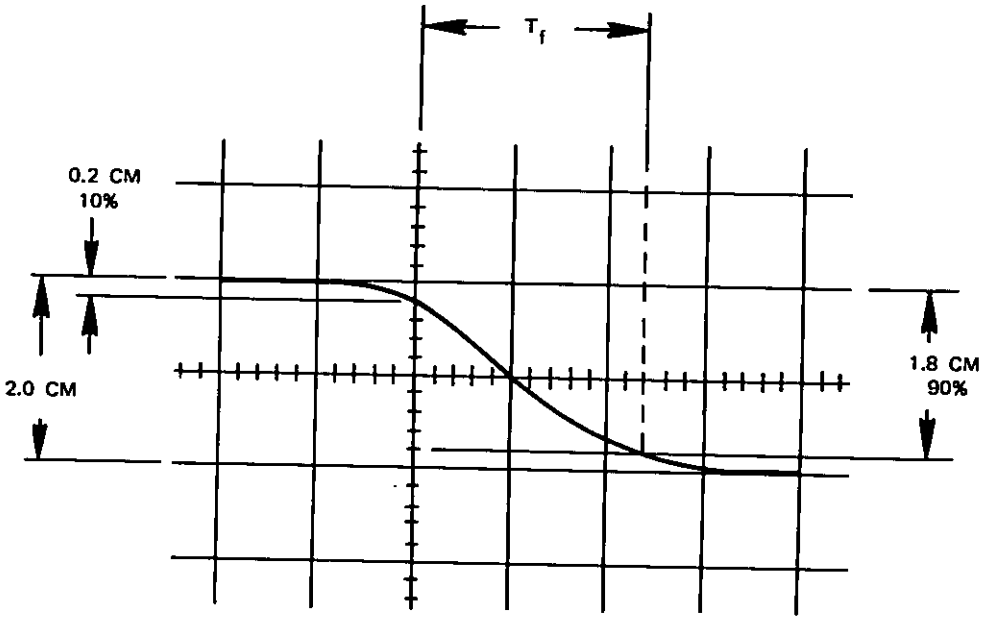


FIGURE 5: Fall time trace

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4.7.3.11.2 Adjust oscilloscope trace to center horizontal line of oscilloscope graticule. Position Actuator Controller Test Panel POWER switch to ON. Monitor oscilloscope voltage waveform. Voltage shall increase to -22 to -26 volts for approximately 2 seconds then return to -1.0 volts maximum. Position Actuator Controller Test Panel POWER switch to OFF.

4.7.3.11.3 Position electronic counter controls as follows:

Control	Position
MANUAL GATE	CLOSED
DISPLAY TIME	FULL CW
FUNCTION SELECTOR	TIME INTERVAL
TIME UNIT	SECONDS, 0.1 STD. FREQ. COUNTED
TRIGGER INPUT	COM
TRIGGER SLOPE (LEFT SW)	(-)
TRIGGER SLOPE (RIGHT SW)	(+)
TRIGGER LEVEL VOLTS (LEFT)	-3
TRIGGER LEVEL VOLTS (RIGHT)	-3

4.7.3.11.4 Connect electronic counter START input to Actuator Controller Test Panel PITCH INHIBIT jack. Position Actuator Controller Test Panel POWER switch to ON. Time indicated by electronic counter shall be 1.8 to 2.2 seconds.

4.7.3.11.5 Disconnect oscilloscope and electronic counter.

4.7.3.12 Low voltage circuit.

4.7.3.12.1 Connect digital voltmeter to actuator controller test panel -40V jack. Decrease -40 volt regulated power supply voltage until actuator controller test panel +40 VOLT CURRENT ammeter indication drops to less than 0.5 amperes. Indication on digital voltmeter shall be -22.5 to -28.5 vdc.

4.7.3.12.2 Increase -40 volts regulated power supply voltage until actuator controller test panel +40 VOLT CURRENT ammeter indication increases to greater than 1.0 ampere. Indication on digital voltmeter shall be -23.0 to -29.0 vdc.

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4.7.3.12.3 Position all power unit test set POWER switches to OFF. Record time indicated on RUNNING TIME meter.

4.7.4 Visual inspections.

4.7.4.1 Identification marking. (Major)

4.7.4.1.1 The power unit shall be identified permanently in accordance with the applicable documents listed in DL 2539015.

4.7.4.2 Interchangeability. (Major)

4.7.4.2.1 Connector receptacle shall be correct and capable of accepting matching unit without resorting to selection, modification, or fitting. (see DWG 2539015 or 2557500)

4.7.4.2.2 Diameter dimension shall be 11.984 inches maximum. (see DWG. 2539016 or 2557500)

4.7.2.3 Thickness dimension shall be 2.510 inches maximum. (see DWG. 2539016 or 2557500)

4.7.5 Process verification.

4.7.5.1 Criteria stipulated in this paragraph and it subparagraphs are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.7.5.2 Lacing. (Minor)

4.7.5.2.1 Lacing and cable configuration shall be in accordance with applicable drawings of DL 2539015 or DL 2557500. Specific method of verifying adequacy of cable lacing shall include visual and dimensional verification.

4.7.5.3 Soldering. (Minor)

4.7.5.3.1 Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.7.5.4 Cleanliness. (Major)

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4.7.5.4.1 Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.7.6 Packaging, packing and marking inspections.

4.7.6.1 Packaging. (Minor)

4.7.6.1.1 Level A packagin shall comply with DL 1330425. If Level C packaging is used, packaging shall comply with commercial practice.

4.7.6.2 Packing. (Major)

4.7.6.2.1 Level A packing shall be in accordance with MIL-P-116, Method III. If Level C packing is used, the power unit shall be protected against damage during direct shipment from supplier to first receiving activity for immediate use.

4.7.6.3 Marking. (Major)

4.7.6.3.1 Destination and other shipment markings shall be correct, legible, and in accordance with DL 1330425.

4.8 Rejection criteria. Failure of the power unit to meet any of the requriements and tests of this specification shall be considered cause for rejection of the unit, and in the case of periodic production tests, of the lot from which the sample was drawn.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2(c) and 6.4).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for Level A shipment shall be in accordance with instructions contained in applicable documents listing in DL 1330425.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2(d) and 6.4).

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5.2.1 Level A. Multiples of the power unit, preserved and packaged in accordance with 5.1.1, shall be packed in accordance with MIL-P-116, Method III.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The power unit, preserved and packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal marking. In addition to markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with DL 1330425. Any additional marking shall be in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The power unit is intended for use in the Torpedo Mk 46 Mod 1.

6.1.2 Description. The power unit consists of semiconductor devices in circuitry designed to provide and regulate direct and alternating current voltages within close tolerances under various conditions of input voltages and output load voltages and output load variations. The power unit also accepts control signals from associated components and modifies them for processing by steering control components. The Type I power unit consists of one conventionally wired module made of a cylindrical metal casting. The Type II power unit comprises two cylindrical modules. One of the modules is conventionally wired and its components are mounted on a cylindrical metal casting. The other module uses printed-circuit wiring and its components are mounted on a cylindrical plastic plate.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type required (see 1.2).

c. If the requirement for a first article inspection is dropped (see 3.1).

d. Required level of preservation and packaging (see 5.1).

e. Required level of packing (see 5.1).

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f. Assigned activity for first article and periodic production tests.

g. Any exceptions to this specification.

h. All additional samples required shall be provided and inspected at the expense of the supplier when such additional samples are required because of failure of any sample(s) to pass the prescribed inspection.

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.0(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENT	APPLICABLE DID
4.7.1.2	Reports, Tests, Specification Appendant	DI-T-20602

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, ASMDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions. For the purpose of this specification, the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Nonoperating: The condition of the unistalled power unit when stored, transported, or handled (see 5.1.1).

6.3.1.2 Ready, nonoperating. The condition of the power unit when assembled with the end item, transported and handled.

6.3.1.3 Ready. The condition of the power unit when fully assembled with the end item and installed on the lauching mechanism prior to operating.

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6.3.1.4 Operating. The condition of the power unit when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (Hz).

6.3.3 Shock. Shock is an abrupt change in the position of a body, resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and in the form of a shock pulse approximately a one-half sine wave. The time duration shall be measured between two points where two lines, tangent a maximum slope, intersect the zero g line.

6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body, resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature. The temerature of the environment to which the power unit may be exposed, expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. The pressure of the environment to which the power unit may be exposed, expressed in pounds per square inch gage (psig).

6.3.7 Humidity. The relative humidity of the environment to which the power unit is exposed, expressed as a percent.

6.3.8 Defective unit. A unit shall be considered defective if it does not meet all quality conformance examinations or tests specified, or deviations are found through visual examination after environmental exposure.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

a. Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

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b. Level B. Not applicable.

c. Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first received activity.

6.5 Storage life. The power unit, preserved and packaged in accordance with Section 5, is intended to withstand an unattended storage life of five years.

6.6 Changes for previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to to extensiveness of the changes.

Preparing Activity:
NAVY - OS

Project Number:
1355-N399

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

POWER UNIT

DL 2539015 (Type I) or DL 2557500 (Type II)

(For Torpedo MK 46 Mod 1)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

POWER UNIT DL 2539015 (Type I) or DL 2557500 (Type II)

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the power unit bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the power unit. Unless otherwise noted, the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limits column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limits column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment	4.7.2	In Calibration	_____
30.2 Electrical tests		Type I/II	_____
30.2.1 Initial test setup			
Pins UU and	4.7.3.1.1		
F		(b) 0.5 ohms max	_____
H		0.5 ohms max	_____

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	Reference	Limits	Reading
30.2.1 (Continued)			
J		0.5 ohms max	_____
U		0.5 ohms max	_____
Y		0.5 ohms max	_____
SS		0.5 ohms max	_____
VV		0.5 ohms max	_____
Casting and pin	4.7.3.1.1		
A		(d)Greater than 500K	_____
B		Greater than 500K	_____
R		Greater than 500K	_____
S		Greater than 500K	_____
T		Greater than 500K	_____
HH		Greater than 500K	_____
TT		Greater than 500K	_____
UU		Greater than 500K	_____
WW		Greater than 500K	_____
XX		Greater than 500K	_____
Pins HH and XX	4.7.3.1.1		
		(e)Greater than 500 ohms	_____
Pins TT and WW	4.7.3.1.1		
		(f)Greater than 1K	_____
Running time meter	4.7.3.1.3	zero meter	_____

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		Reference	Limits	Reading
30.2.2	+36 vdc and -36 vdc Input (maximum load)			
	-40 volt current	4.7.3.2.3	1.5 to 5.0 amps	_____
	+40 volt current	4.7.3.2.4	0.5 to 3.0 amps	_____
	+5V	4.7.3.2.5	+4.75 to +5.25 vdc	_____
	-15V		-14.5 to -15.5 vdc	_____
	-30V		-29.5 to -31.2 vdc	_____
	-30VG		-28.9 to -30.9 vdc	_____
	+90V (upper)		+90.0 to +100.0 vdc	_____
	+15V		+14.5 to +15.5 vdc	_____
	+30V		+29.4 to +30.6 vdc	_____
	+90V (lower)		+90.0 to +100.0 vdc	_____
30.2.3	4.8 KC supply (maximum load)			
	4.8 KC IN Ø			
	Voltage	4.7.3.3.1	25.5 to 26.5 vrms	_____
	Frequency	4.7.3.3.2	4770 to 4830 Hz	_____
	Distortion	4.7.3.3.4	3% max	_____
30.2.4	+46 vdc and -46 vdc input (minimum load)			
	Q1508	4.7.3.4.4	No oscillations	_____
	+5V	4.7.3.4.6	+4.75 to +5.25 vdc	_____
	-15V		-14.5 to -15.5 vdc	_____
	-30V		-29.5 to -30.8 vdc	_____

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30.2.4 (Continued)	Reference	Limits	Reading
-30VG		-28.9 to -30.5 vdc	_____
+90V (upper)		+90.0 to +100.0 vdc	_____
+15V		+14.5 to +15.5 vdc	_____
+30V		+29.4 to +30.6 vdc	_____
+90V		+90.0 to +100.0 vdc	_____
30.2.5 4.8 KC supply (minimum load)			
4.8 KN IN Ø			
Voltage	4.7.3.5.1	25.5 to 26.5 vrms	_____
Frequency	4.7.3.5.2	4770 to 4830 Hz	_____
Distortion	4.7.3.5.4	3% max	_____
30.2.6 Exploder arm			
EXP-ARM	4.7.3.6.3	-0.1 to +0.4 vdc	_____
EXP-ARM to -40V	4.7.3.6.4	0.0 to -1.5 vdc	_____
EXP-ARM		Variations ≤ 0.5 vdc	_____
EXP-ARM to -40V	4.7.3.6.5	0.0 to -1.5 vdc	_____
30.2.7 Voltage regulator			
SHUNT FIELD	4.7.3.7.3	0.0 to +1.5 vdc	_____
SHUNT FIELD	4.7.3.7.4	+30.0 to +40.0 vdc	_____
-40V	4.7.3.7.5	-40.0 to -41.2 vdc	_____
SHUNT FIELD	4.7.3.7.6	-60.0 to -42.- vdc	_____
Running time meter	4.7.3.7.7	Minutes	_____

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		Reference	Limits	Reading
30.2.8	Actuator controller			
	Running time meter	4.7.3.8.2	Minutes	_____
	CONTINUITY		(Lamp)	_____
	+40V current	4.7.3.8.5	0.85 to 1.55 amps	_____
	-40 current		1.65 to 1.95 amps	_____
	UP, RIGHT, CW lamps	4.7.3.8.6	Proper state	_____
	Function switch	4.7.3.8.7		
	1		+0.20 to -0.10 vdc	_____
	2		-21.8 to -25.8 vdc	_____
	3		+0.10 to -0.20 vdc	_____
	4		+34.0 to +37.0 vdc	_____
	5		-34.0 to -37.0 vdc	_____
	6		+0.20 to -0.10 vdc	_____
	UP, RIGHT, CCW LAMP	4.7.3.8.8	Proper state	_____
	+40V current	4.7.3.8.9	0.65 to 1.35 amps	_____
	-40V		1.95 to 2.25 amps	_____
	Function switch	4.7.3.8.10		
	2		-34.0 to -37.0 vdc	_____
	4		+21.8 to +25.8 vdc	_____
	DN, LEFT, CCW lamps	4.7.3.8.11	Proper state	_____
	+40V current	4.7.3.8.12	1.40 to 2.10 amps	_____
	-40V current		1.25 to 1.55 amps	_____

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		Reference	Limits	Reading
30.2.8	(Continued)			
	Function switch	4.7.3.8.13		
	1		+21.8 to +25.8 vdc	_____
	2		+0.10 to -0.20 vdc	_____
	3		-34.0 to -37.0 vdc	_____
	4		+0.20 to -0.10 vdc	_____
	5		+0.10 to -0.20 vdc	_____
	6		+34.0 to +37.0 vdc	_____
	DN, LEFT, CW lamps	4.7.3.8.14	Proper state	_____
	+40V current	4.7.3.8.15	1.65 to 2.35 amps	_____
	-40V current		0.95 to 1.25 amps	_____
	Function switch	4.7.3.8.16		
	1		+34.0 to +37.0 vdc	_____
	3		-21.8 to -25.8 vdc	_____
30.2.9	Controller oscillator	4.7.3.9.2		
	V1		-1.1 vdc max	_____
	V2		-22.5 to -24.5 vdc	_____
	T1		3.6 to 5.2 cm	_____
	T2	4.7.3.9.3	6.4 to 7.0 cm	_____
30.2.10	Pulse current			
	Pulse amplitude	4.7.3.10.3		
	1		+16 to +25 vdc	_____
	4		+16 to +25 vdc	_____
	6		+16 to +25 vdc	_____

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		Reference	Limits	Reading
30.2.10 (Continued)				
	Pulse amplitude	4.7.3.10.4		
	2		-16 to -25 vdc	_____
	3		-16 to -25 vdc	_____
	5		-16 to -25 vdc	_____
	Fall time	4.7.3.10.6		
	1		25 usec max	_____
	2		50 usec max	_____
	3		50 usec max	_____
	4		25 usec max	_____
	5		50 usec max	_____
	6		25 usec max	_____
30.2.11	Pitch Inhibit			
	Voltage	4.7.3.11.2	-22 to -26 vdc	_____
			0.0 to -1.0 vdc	_____
	Delay	4.7.3.11.4	1.8 to 2.2 sec	_____
30.2.12	Low voltage circuit			
	-40V Decrease	4.7.3.12.1	-22.5 to -28.5 vdc	_____
	-40V Increase	4.7.3.12.2	-23.0 to -29.0 vdc	_____
	Running time meter	4.7.3.12.3	Minutes	_____
30.3	Visual inspections			
30.3.1	Identification marking	4.7.4.1.1		_____

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		Reference	Limits	Reading
30.3.2	Interchangeability			
	Connectors	4.7.4.2.1		
	Diameter	4.7.4.2.2	11.984 inches max	_____
	Thickness	4.7.4.2.3	2.510 inches max	_____
30.4	Process verification			
30.4.1	Lacing	4.7.5.2.1		_____
30.4.2	Soldering	4.7.5.3.1		_____
30.4.3	Cleanliness	4.7.5.4.1		_____
30.5	Packaging, packing and marking instructions			
30.5.1	Packaging	4.7.6.1.1		_____
30.5.2	Packing	4.7.6.2.1		_____
30.5.3	Marking	4.7.6.3.1		_____

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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